ASSIGNMENT COVER SHEET

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MIT653 – Assessment 2-1

Literature Review Report

The Effectiveness of Generative AI in Creating Personalised Video and Audio Content

Abstract

Digital content creation has been drastically altered by generative artificial intelligence (GenAI), especially in the areas of speech, music, and video production. With less technical expertise and production time, users can now create customized digital content thanks to recent developments in text-to-video, text-to-music, and speech synthesis.

Effectiveness, however, cannot be assessed solely by practical results. Other factors that need to be taken into account include content quality, personalization, production efficiency, authenticity, copyright, privacy, misinformation, bias, and human creativity.

This review of the literature looks at recent studies in three primary areas: production efficiency, ethical issues, and content quality and personalization. According to the literature, GenAI can increase production speed, scalability, and accessibility, but there are still issues with consistency, controllability, semantic accuracy, authenticity, and ethical governance. Therefore, the review contends that human-AI collaboration, which combines automated generation with human creative direction, quality control, and ethical judgment, is the best way to understand effective use of GenAI.

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1. Introduction & Background

For the production of digital content, generative artificial intelligence has grown in importance. Generative AI can create new text, images, music, speech, and video, in contrast to conventional AI systems that primarily analyze or categorize existing data. Users can now create digital media with relatively simple instructions thanks to recent developments in deep learning and generative models. Because content creation has historically required specialized technical skills, costly software, and a significant amount of time, this development has important implications for creative industries.

The production of audio and video is especially relevant to generative AI. While text-to-music and speech-generation systems can produce audio in response to user commands, text-to-video systems can convert textual descriptions into video sequences. Models that show quick development in these areas include Make-A-Video, Imagen Video, VideoPoet, MusicLM, MusicGen, and Voicebox. Advertising, education, entertainment, social media, and customized communication are all made possible by these technologies.

The evolution of these technologies also modifies the function of human creators. AI is increasingly being used by creators to create original content, edit pre-existing content, and automate tedious tasks rather than creating each component by hand. This has the potential to boost productivity and lower production costs. However, generated audio may raise questions about voice identity, authenticity, and unauthorized imitation, while generated video may exhibit temporal irregularities or unrealistic physical behaviour.

The ethical aspect is just as significant. Despite worries about fraud, deepfakes, and copyright infringement, Barnett's systematic review of generative-audio research revealed that negative effects were discussed far less frequently than positive applications. This implies that technology can advance more quickly than methodical analysis of its social implications.

Thus, from the standpoints of quality and personalisation, production efficiency, and ethical responsibility, this review investigates the efficacy of generative AI in producing customised audio and video content.

2. Purpose & Aims of the Literature Review

This review of the literature aims to critically analyze studies on generative AI for the production of customized audio and video content. Three research questions are addressed in the review:

RQ1: What effects does generative AI have on the relevance, quality, realism, and inventiveness of customized audio and video content?

RQ2: How can generative AI enhance digital content production's scalability, accessibility, and efficiency?

RQ3: In terms of authenticity, copyright, privacy, misinformation, bias, and human creativity, what ethical issues are raised by AI-generated audio and video?

The review is not a summary of single papers but a synthesis of findings across studies. It also identifies common trends, strengths, limitations and research gaps.

3. Literature Review Process

The literature search was mainly based on research published between 2020 and 2026 as per the assessment requirement to prioritize recent literature. We used academic and reputable online sources. search terms were

- "generative AI"
- "AI-generated video"
- "text-to-video generation"
- "generative audio"
- "AI music generation"
- "speech generation"
- "voice cloning"
- "personalised digital content"
- "content quality"
- "production efficiency"
- "deepfake"
- "AI ethics"
- "copyright"
- "synthetic media"

Four themes—technological advancement, content quality and personalisation, production efficiency, and ethical issues—were used to categorise the literature. Peer-reviewed journal articles and established conference proceedings were prioritised. Technical papers outlining significant generative models that clearly illustrated the strengths and weaknesses of existing systems were included.

The chosen studies were compared using a literature matrix according to their focus, methodology, main conclusions, limitations, and applicability to the research questions. Because they offer fair coverage of video, audio, efficiency, quality, and ethics, ten important studies were chosen for the matrix. The final reference list may include additional reliable scholarly and internet sources to bolster background, industry, and regulatory claims.

3.1 Literature Review Matrix

Table 1 summarises ten key studies used as the foundation for the thematic synthesis.

No.	Study	Focus	Method	Key Findings	Limitations / Gap	Relevance
1	Singer et al. (2022)	Make-A-Video / video	Model development and evaluation	Demonstrates text-to-video generation and improved video synthesis capability.	Quality and temporal consistency remain challenging.	Video quality
2	Ho et al. (2022)	Imagen Video / video	Diffusion-model development and evaluation	Demonstrates high-definition video generation with strong controllability.	Computational demands and generation limitations.	Video quality
3	Kondratyuk et al. (2024)	VideoPoet / multimodal media	Multimodal model development and evaluation	Supports generation across text, image, video, and	Multimodal evaluation and control remain	Personalisation

				audio modalities.	challenging.	
4	Agostinelli et al. (2023)	MusicLM / music	Generative model and evaluation	Generates music from textual descriptions and supports conditioning.	Fine-grained controllability remains limited.	Personalised audio
5	Copet et al. (2023)	MusicGen / music	Model development and human/automatic evaluation	Provides controllable text-to-music generation and improved generation quality.	Human refinement can still be required.	Quality and control
6	Le et al. (2023)	Voicebox / speech	Experimental model evaluation	Demonstrates scalable and efficient multilingual speech generation.	Voice misuse and safety require attention.	Production efficiency
7	Barnett (2023)	Generative audio ethics	Systematic literature review	Reviews 884 generative-audio papers and identifies under-discussed negative impacts.	Ethical research is less developed than technical research.	Ethics

8	Mirsky & Lee (2021)	Deepfakes	Survey	Shows the growing challenge of creating and detecting synthetic media.	Generation and detection create an ongoing arms race.	Authenticity
9	Hagendorff (2024)	Generative AI ethics	Comprehensive scoping review	Maps a broad range of ethical issues across generative AI.	Practical governance mechanisms need further development.	Ethics
10	Amankwah-Amoah et al. (2024)	Creative industries	Conceptual / research agenda	Highlights productivity and innovation benefits alongside disruption to creative work.	More empirical evidence is needed on long-term labour effects.	Efficiency and human labour

Table 1. Literature review matrix of ten key studies.

4. Main Body of the Report

The matrix shows a distinct progression from technical proficiency to practical efficacy and responsible use issues. The literature can be summarized in terms of research gaps, production efficiency, ethical issues, and content quality and personalization.

4.1 Generative AI for Video and Audio Content Creation

Recent studies show that generative audio and video are developing quickly. VideoPoet expands the method to multimodal generation, while Make-A-Video and Imagen Video demonstrate how textual descriptions can be transformed into visual sequences. Voicebox shows scalable speech generation in audio, while MusicLM and MusicGen show text-conditioned music generation.

All of the research indicates that GenAI is shifting from producing single-purpose content to producing more adaptable multimodal content [11]–[15]. Technical proficiency does not, however, guarantee that produced content is appropriate for use in a professional setting. Quality, consistency, controllability, relevance, and the degree of human intervention needed all affect effectiveness.

4.2 Content Quality and Personalisation

A key indicator of the efficacy of generative AI is content quality. While audio models have increased naturalness and controllability, video models have improved visual fidelity and semantic alignment. However, quality is multifaceted. A video may have temporal or physical inconsistencies throughout a sequence even though it looks realistic in individual frames. In a similar vein, even though generated audio is technically clear, it might not convey the intended identity, style, or emotion.

Another requirement brought about by personalisation is that the output must be appropriate for the user's intended use. For instance, a customised instructional video must be pertinent to the learner, and audience preferences must be reflected in customised marketing content. Thus, in addition to technical realism, relevance and controllability should be assessed [14], [15], [19].

The matrix also demonstrates a discrepancy between human judgment and automated metrics. Technical metrics should be combined with user evaluations of usefulness, relevance, realism, and reliability in future assessments.

4.3 Production Efficiency and Accessibility

One of GenAI's greatest potential benefits is production efficiency. Planning, recording, editing, sound design, visual effects, and post-production are all part of traditional video and audio production. Parts of these processes can be automated by generative systems, which also lessen the technical know-how needed for initial content creation.

Multimodal video systems, Voicebox, MusicLM, and MusicGen are examples of how AI can increase the scalability of content creation. Without creating each version by hand, users can create multiple versions of content for various audiences. This is especially pertinent to social media, education, advertising, and multilingual communication [16]–[20].

However, total production efficiency should not be equated with generation speed. There may still be a need for human review, editing, fact-checking, copyright checks, and ethical approval. Therefore, the entire workflow from the first prompt to the final approved content should be taken into account for the true efficiency gain.

4.4 Ethical Challenges

The ethical literature lists a number of dangers connected to artificial audio and video.

First, two of the biggest issues are authenticity and deepfakes. Audiences may find it challenging to discern between generated content and real events as synthetic media become more realistic. False information, fraud, impersonation, and reputational damage can all be facilitated by deepfakes. The ongoing advancements in generation technology also pose a challenge to detection systems.

Second, ownership and copyright are still unsolved problems. Large collections of already-existing media may be used to train generative systems, raising issues with training data, authorship, ownership, and original creators' rights. These concerns are particularly pertinent to voice production and music.

Third, because a person's voice can serve as a distinguishing feature of their identity, voice cloning raises privacy and consent issues. Therefore, consent, openness, and systems that let people manage how their voice is used are necessary for responsible systems.

Fourth, biases from training data can be replicated by generative systems. Personalisation may inadvertently create unfair representations or perpetuate stereotypes. This is especially crucial for public communication, advertising, and education.

Finally, GenAI might alter the nature of creative work. Automation may change jobs and professional roles, but it can also cut down on repetitive production tasks. According to the literature, human-AI cooperation might be more sustainable than replacing human creators entirely. Transparency, accountability, fairness, human oversight, privacy protection, and responsible governance are crucial when deploying generative AI systems, according to recent ethical guidelines and regulatory frameworks [21]–[25].

4.5 Critical Analysis and Research Gaps

There are a number of significant gaps in the literature. First, research on adverse effects is less developed than technical development. Despite issues like fraud, deepfakes, and copyright infringement, Barnett's review of 884 generative-audio papers revealed that less than 10% addressed negative effects. This suggests that capability and impact research are not balanced.

Second, the term "content quality" for generative media has no universally recognised definition. Technical metrics may not capture usefulness, cultural appropriateness, trust, or alignment with user intentions, but they can measure visual or audio qualities.

Third, systems are evaluated in controlled environments in a lot of model studies. Editing, human approval, copyright checks, and audience expectations are all part of real production. Therefore, more studies should assess GenAI in practical production processes.

Fourth, more empirical study is required to determine how GenAI will affect creative professionals in the long run. While new roles in prompting, editing, creative direction, and quality assurance may be created, some tasks may be replaced by technology.

Finally, there is still fragmentation in ethical governance. It is insufficient to just identify ethical risks; real-world evaluations of practical mechanisms like provenance, content labelling, consent systems, copyright management, and human review are required.

According to the literature, a multifaceted framework that incorporates technical quality, production efficiency, personalisation, user satisfaction, authenticity, ethical risk, and human oversight should be used to evaluate the efficacy of GenAI.

5. Conclusions

The production of customised audio and video has shown great promise thanks to generative AI. The chosen literature demonstrates advancements in production efficiency, scalability, multimodal capability, and generation quality. Consistency, controllability, authenticity, copyright, privacy, bias, and creative labour are among the ongoing issues highlighted by the studies.

Thus, the literature is in favour of a balanced view. The ability of GenAI to produce content quickly and realistically should not be the only criterion for evaluating it. For the content to be used effectively, it must be relevant, reliable, ethically produced, and appropriate for the target audience. For creative direction, quality assurance, contextual judgment, and ethical responsibility, human oversight is still crucial.

Future studies should create evaluation frameworks that incorporate human judgment, ethical standards, and technical metrics. Realistic production settings and the long-term effects of GenAI on creative work should also be investigated.

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